

Dutta E91-013: integrals of the (e,e'p) spectral-function data

Liang Liu

September 10, 2026

Summary of what the author data tables behind the Dutta *et al.* JLab Hall C **E91-013** paper (nucl-ex/0303011, ^{12}C / ^{56}Fe / ^{197}Au quasi-elastic (e,e'p)) integrate to, and on which scale. This document is the write-up that goes with `integrate_dutta.py` (the integrator) and `make_published_vs_table.py` (the published-vs-table figures of section 1); it is generated by `make_report_tex.py` from the same data files as the markdown version (`README.md`).

- Paper source: `papers/nuc1-ex_0303011/` (tex `longpaper2.tex`; every `tex:line` below anchors to it); published figure renders in `papers/nuc1-ex_0303011/figures/`.
- Author data: `data/Dipingkar-dutta-data-prc_figs/` — 14 files, exactly figs 6, 7, 9, 11 of the paper.
- Earlier per-figure report: `report/dutta-e91013-figures.md`. Its normalization reading of figs 9/11 (“ $\approx Z$, full-occupancy scale”) is the old one; the sections after section 1 here supersede it.

Sections 4 (the conventions behind the factor $\frac{1}{2}$ and the positive-half 3D integral; nucleon counts against $T \cdot Z/f_{\text{corr}}$) and 5 (windowed integrals, shell occupancies, caveats) are still to be written.

1 E_m and p_m for C12 and Fe56: each published figure next to its data table

1.1 The four figures, published vs table

Left: the paper’s render (autocropped). Right: the same quantity drawn from the `.dat` files on the paper’s axes — points with the tabulated statistical errors, **no rescaling** (no $\frac{1}{2}$ on the E_m files, no L+R fold on the p_m files; those conventions are the subject of the next sections). The p_m replots use the house colour cycle with fig 6’s marker shapes per Q^2 (fig 7 in print assigns markers differently; the legends identify the sets). Numbers quoted below come from `dutta_published_vs_table.txt`: a per-file ratio to the $Q^2 = 1.8$ reference and a histdiag `describe1d` readout of all 14 files, computed on the tabulated values.

^{12}C missing energy (fig 9). The 16 tabulated points are the published ones: $0.571 \pm 0.005 \text{ MeV}^{-1}$ at $E_m = 17.5 \text{ MeV}$, 0.269 at 22.5, the s-shell bump peaking at 0.077 at 37.5, and three exact zeros below 15 MeV. The IPSM curve exists only in print. The one visible difference is the error bars: the file’s column 4 is statistical (0.8% at the peak), while the published bars at 17.5 and 22.5 MeV are several times larger (pixel-measured in `dutta-e91013-figures.md` §5).

Dutta fig. 9: ^{12}C missing-energy spectral function, $Q^2 = 1.28 \text{ (GeV/c)}^2$

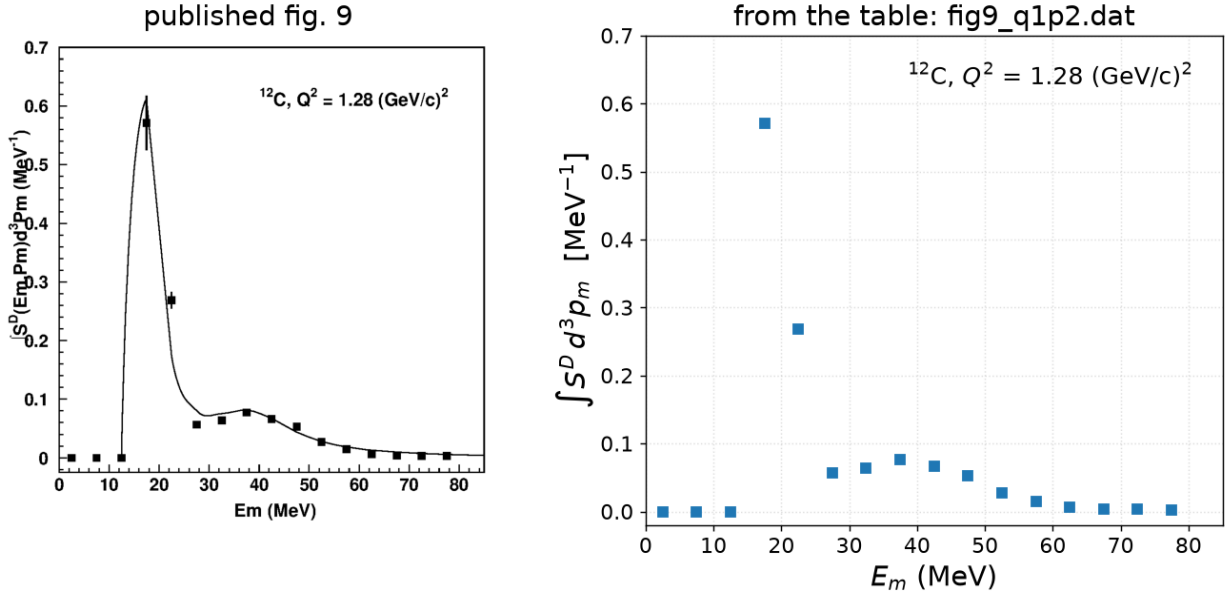


Figure 1: Fig. 9 of the paper (left) and the 16 points of fig9_q1p2.dat (right).

^{12}C missing momentum, p-shell and s-shell windows (fig 6). Three of the four Q^2 sets sit on the published points: relative to the $Q^2 = 1.8$ reference file the bin-wise median ratio is 1.01 ($Q^2 = 1.28$) and 0.97 (3.25) in the p-shell panel, 1.14 and 1.20 in the s-shell panel, with the signed-axis sums equal to within 5–8 %. The $Q^2 = 0.64$ files **do not**: their median ratio to the reference is 1.27 (p-shell) and 1.33 (s-shell) with a p_m -dependent spread (1.07–1.44), whereas in print all four Q^2 coincide within marker size. Those two files are excluded from every integral in this document (open question tracked in `open_questions.md`). The $\ell = 1$ dip at $p_m = 0$ in the p-shell window and the $\ell = 0$ peak in the s-shell window (tex:920–922) are in the tables as printed: in the $Q^2 = 1.28$ file the p-shell dip bin is 0.31 of its peak at $|p_m| = 100 \text{ MeV/c}$.

^{56}Fe missing energy (fig 11). The table reproduces the published points ($0.810 \pm 0.009 \text{ MeV}^{-1}$ at $E_m = 12.5 \text{ MeV}$, then a monotone fall to 0.055 at 77.5; two exact zeros below 10 MeV). The three theory curves (IPSM, Benhar, TIMORA) exist only in print.

^{56}Fe missing momentum (fig 7). All four files coincide as printed: median ratios to the $Q^2 = 1.8$ reference 1.17 (0.64), 1.09 (1.28), 1.08 (3.25), signed-axis sums equal to within 5 % (the caption's normalization). No fig 6-type anomaly here. Bin-wise statistical errors are 1–5 %, largest at $|p_m| = 20$ and 300 MeV/c.

Dutta fig. 6: ^{12}C missing-momentum distributions

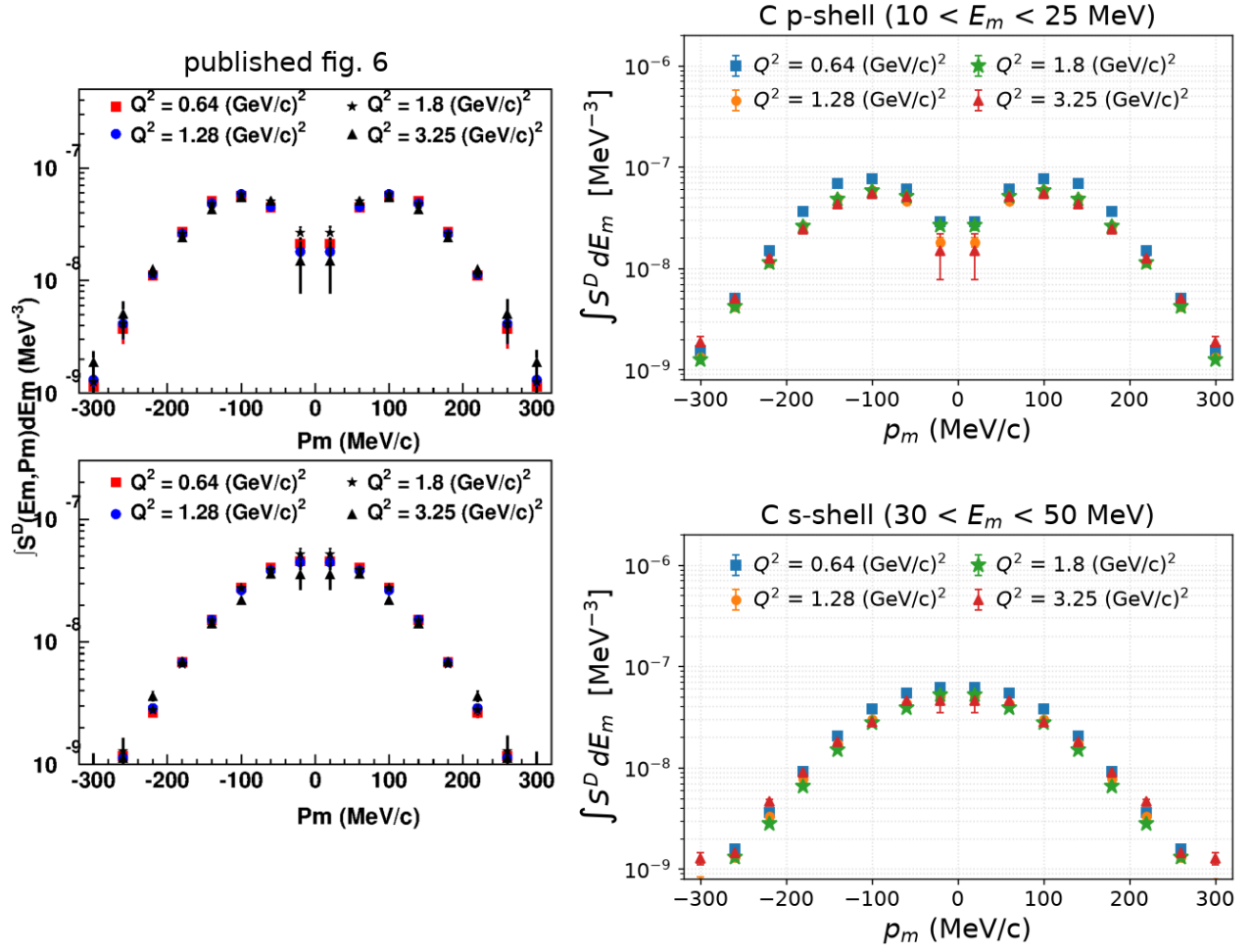


Figure 2: Fig. 6 of the paper (left) and the eight fig6_*.dat files (right), four Q^2 sets per panel.

Dutta fig. 11: ^{56}Fe missing-energy spectral function, $Q^2 = 1.28 \text{ (GeV/c)}^2$

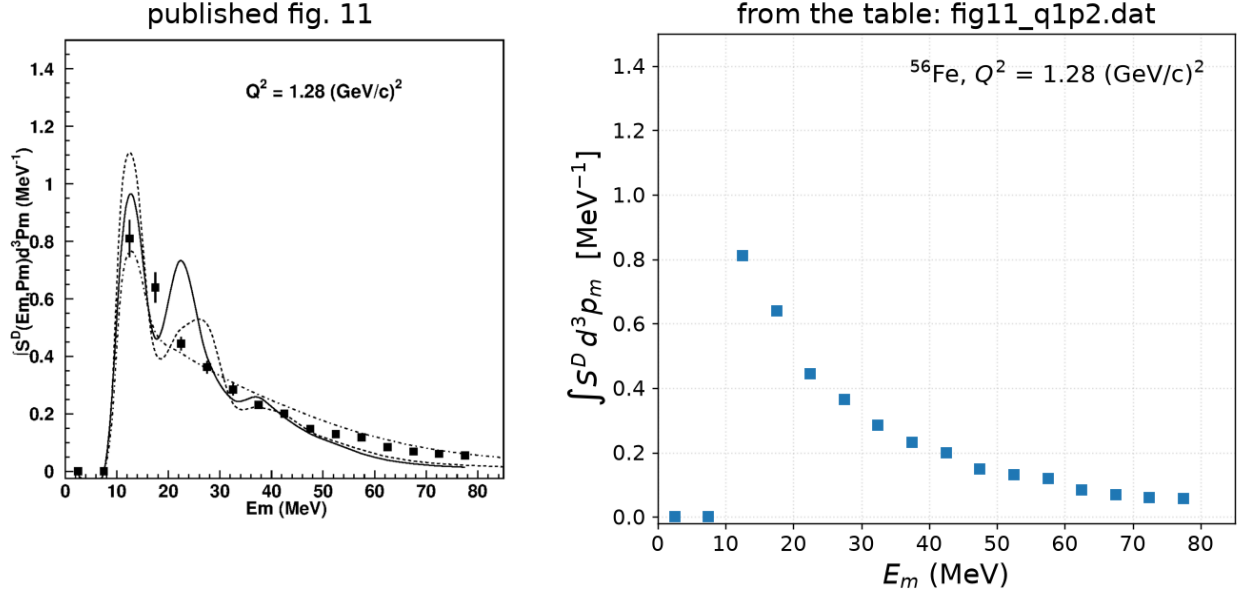


Figure 3: Fig. 11 of the paper (left) and the 16 points of fig11_q1p2.dat (right).

Dutta fig. 7: ^{56}Fe missing-momentum distribution

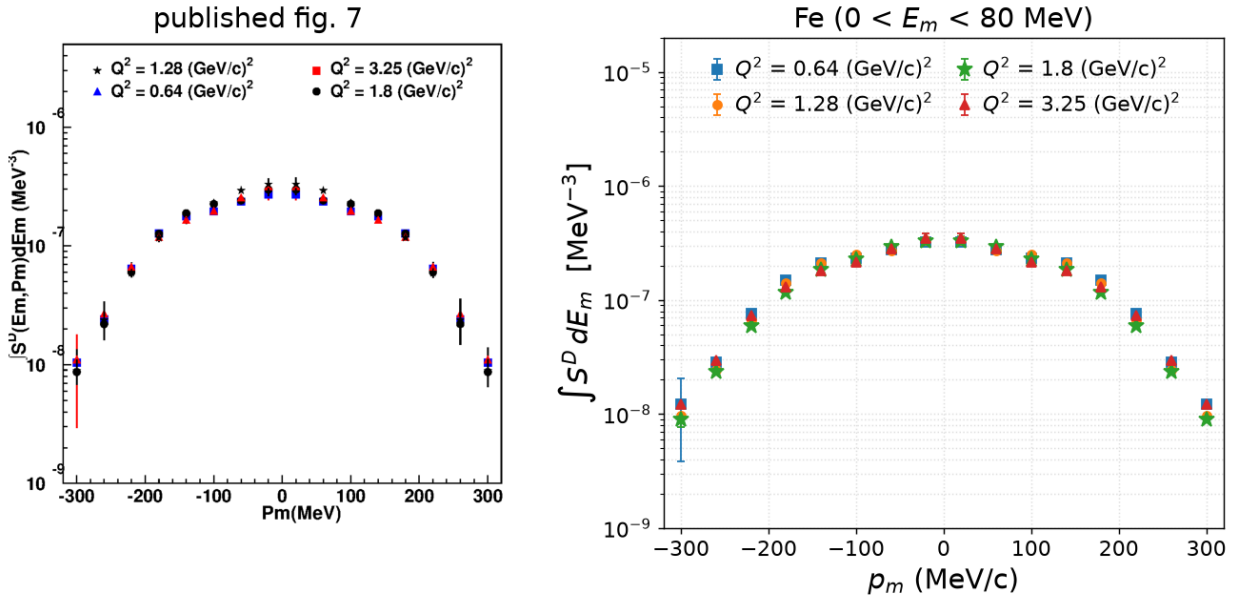


Figure 4: Fig. 7 of the paper (left) and the four fig7_*.dat files (right).

1.2 Reproduce

From the repo root:

```
pixi run python report/dutta-integral/make_published_vs_table.py
# -> figures/dutta_fig{9,6,11,7}_*_published_vs_table.png
#   figures/dutta_published_vs_table.txt
pixi run python report/dutta-integral/make_report_tex.py      # -> dutta-integral.tex
cd report/dutta-integral && pixi run \
  --manifest-path /exp/dune/data/users/liangliu/texenv/pixi.toml \
  tectonic --outdir . dutta-integral.tex                      # -> dutta-integral.pdf
```

The script reads the .dat files directly, uses the house plot style (results/template/plot_style.py) and writes its histdiag readout with results/template/histdiag.py; the published renders come from papers/nucl-ex_0303011/figures/.

2 Missing energy at $Q^2 = 1.28 \text{ (GeV/c)}^2$: the tabulated points and their sums

2.1 ^{12}C (fig 9)

The 16 rows of fig9_q1p2.dat (column 1 = bin centre, column 2 = $S(E_m)$ as plotted):

bin	E_m (MeV)	$S(E_m)$ (MeV $^{-1}$)
1	2.5	0.00000
2	7.5	0.00000
3	12.5	0.00000
4	17.5	0.57130
5	22.5	0.26883
6	27.5	0.05668
7	32.5	0.06362
8	37.5	0.07718
9	42.5	0.06606
10	47.5	0.05315
11	52.5	0.02708
12	57.5	0.01514
13	62.5	0.00608
14	67.5	0.00436
15	72.5	0.00359
16	77.5	0.00297
sum		1.21604

Sums over the 16 points (errors: column 4 in quadrature, statistical only):

quantity	value
$\sum S(E_m)$, plain sum of the tabulated values	1.21604 $\pm$ 0.00578 MeV ⁻¹
$\sum S(E_m) \Delta E$ with $\Delta E = 5$ MeV (the plotted area, 0–80 MeV)	6.0802 $\pm$ 0.0289
$\frac{1}{2} \sum S(E_m) \Delta E$	3.0401 $\pm$ 0.0145

Where the strength sits: the two p-shell bins at 17.5 and 22.5 MeV carry 69.1 % of the sum, the four s-shell bins 30–50 MeV carry 21.4 %, the dip bin at 27.5 MeV 4.7 % and the tail above 50 MeV 4.9 %. The author’s integral for this figure is **3.04**, i.e. half of $\sum S(E_m) \Delta E$; why the plotted area is twice the nucleon count (the signed $-300 \dots + 300$ MeV/c p_m axis) is the subject of section 3.

Reproduce (the plotted sum and its half are the **plotted sum** and N columns):

```
pixi run python report/dutta-integral/integrate_dutta.py --files fig9_q1p2
```

2.2 ⁵⁶Fe (fig 11)

The 16 rows of `fig11_q1p2.dat` (column 1 = bin centre, column 2 = $S(E_m)$ as plotted):

bin	E_m (MeV)	$S(E_m)$ (MeV ⁻¹)
1	2.5	0.00000
2	7.5	0.00000
3	12.5	0.80983
4	17.5	0.63965
5	22.5	0.44406
6	27.5	0.36284
7	32.5	0.28453
8	37.5	0.23175
9	42.5	0.20011
10	47.5	0.14872
11	52.5	0.13002
12	57.5	0.11820
13	62.5	0.08474
14	67.5	0.06965
15	72.5	0.06070
16	77.5	0.05527
sum		3.64006

Sums over the 16 points (errors: column 4 in quadrature, statistical only):

quantity	value
$\sum S(E_m)$, plain sum of the tabulated values	3.64006 $\pm$ 0.01579 MeV ⁻¹
$\sum S(E_m) \Delta E$ with $\Delta E = 5$ MeV (the plotted area, 0–80 MeV)	18.2003 $\pm$ 0.0790
$\frac{1}{2} \sum S(E_m) \Delta E$	9.1001 $\pm$ 0.0395

Where the strength sits: the three bins 10–25 MeV carry 52.0 % of the sum, 25–50 MeV 33.7 %, and the tail 50–80 MeV 14.2 % (the last bin alone 1.5 %), against 4.9 % above 50 MeV for carbon.

No author-quoted integral exists for this figure; the same $\frac{1}{2}$ convention as for fig 9 gives 9.10, to be cross-checked against the fig 7 momentum-distribution integral in section 4.

Reproduce:

```
pixi run python report/dutta-integral/integrate_dutta.py --files fig11_q1p2
```

3 Missing momentum at $Q^2 = 1.28 \text{ (GeV/c)}^2$: the tabulated points and their integrals

The p_m files are tabulated on the signed axis, 16 bins of $\Delta p = 40 \text{ MeV/c}$ centred at $-300 \dots +300 \text{ MeV/c}$, and every file is exactly left-right symmetric, $y(-p_m) \equiv y(+p_m)$ to full precision. Two integrals are quoted per file:

- **the plotted area** $\sum S(p_m) \Delta p$ over all 16 signed bins (MeV^{-2}) — the quantity the paper’s rescale-to- $Q^2 = 1.8$ caption equalizes;
- **the 3D integral** $N = 4\pi \sum_{p_m > 0} S(p_m) p_m^2 \Delta p$ over the **positive half only** (p_m at the bin centre, rectangle rule), dimensionless — the number of protons in the file’s E_m window. Summing both halves would count each $|p_m|$ twice, since the files are symmetrized.

Errors are column 4 in quadrature (statistical only). All numbers are the **plotted sum** / N columns of `integrate_dutta.py`.

3.1 ^{12}C (fig 6, p-shell and s-shell windows)

The 16 rows of `fig6_top_q1p2.dat` (p-shell window) and `fig6_bot_q1p2.dat` (s-shell window), column 1 = bin centre, column 2 = $S(p_m)$ as plotted, with the per-bin 3D weights ($\Delta p = 40 \text{ MeV/c}$, p_m at the bin centre). As for iron, the 2π column summed over all 16 signed bins is the 3D integral N and the 4π column over all 16 bins is twice that.

p-shell window, $10 < E_m < 25$ MeV.

bin	p_m (MeV/c)	$S(p_m)$, p-shell (MeV ⁻³)	$4\pi S(p_m) p_m^2 \Delta p$	$2\pi S(p_m) p_m^2 \Delta p$
1	-300	1.31822×10^{-9}	0.0596	0.0298
2	-260	4.15417×10^{-9}	0.1412	0.0706
3	-220	1.13985×10^{-8}	0.2773	0.1387
4	-180	2.61633×10^{-8}	0.4261	0.2130
5	-140	4.86048×10^{-8}	0.4789	0.2394
6	-100	5.89047×10^{-8}	0.2961	0.1480
7	-60	4.55982×10^{-8}	0.0825	0.0413
8	-20	1.81916×10^{-8}	0.0037	0.0018
9	+20	1.81916×10^{-8}	0.0037	0.0018
10	+60	4.55982×10^{-8}	0.0825	0.0413
11	+100	5.89047×10^{-8}	0.2961	0.1480
12	+140	4.86048×10^{-8}	0.4789	0.2394
13	+180	2.61633×10^{-8}	0.4261	0.2130
14	+220	1.13985×10^{-8}	0.2773	0.1387
15	+260	4.15417×10^{-9}	0.1412	0.0706
16	+300	1.31822×10^{-9}	0.0596	0.0298
sum, all 16 bins		4.28667×10^{-7}	3.5306	1.7653

s-shell window, $30 < E_m < 50$ MeV.

bin	p_m (MeV/c)	$S(p_m)$, s-shell (MeV ⁻³)	$4\pi S(p_m) p_m^2 \Delta p$	$2\pi S(p_m) p_m^2 \Delta p$
1	-300	7.38319×10^{-10}	0.0334	0.0167
2	-260	1.29075×10^{-9}	0.0439	0.0219
3	-220	3.29055×10^{-9}	0.0801	0.0400
4	-180	7.72794×10^{-9}	0.1259	0.0629
5	-140	1.72738×10^{-8}	0.1702	0.0851
6	-100	3.00671×10^{-8}	0.1511	0.0756
7	-60	4.34589×10^{-8}	0.0786	0.0393
8	-20	5.10294×10^{-8}	0.0103	0.0051
9	+20	5.10294×10^{-8}	0.0103	0.0051
10	+60	4.34589×10^{-8}	0.0786	0.0393
11	+100	3.00671×10^{-8}	0.1511	0.0756
12	+140	1.72738×10^{-8}	0.1702	0.0851
13	+180	7.72794×10^{-9}	0.1259	0.0629
14	+220	3.29055×10^{-9}	0.0801	0.0400
15	+260	1.29075×10^{-9}	0.0439	0.0219
16	+300	7.38319×10^{-10}	0.0334	0.0167
sum, all 16 bins		3.09754×10^{-7}	1.3868	0.6934

quantity	p-shell window	s-shell window
$\sum S(p_m)$, plain sum (MeV ⁻³)	$4.28667 \times 10^{-7} \pm 4.3 \times 10^{-9}$	$3.09754 \times 10^{-7} \pm 3.8 \times 10^{-9}$
$\sum S(p_m) \Delta p$, signed axis (MeV ⁻²)	$1.7147 \times 10^{-5} \pm 1.7 \times 10^{-7}$	$1.2390 \times 10^{-5} \pm 1.5 \times 10^{-7}$
$4\pi \sum_{p_m > 0} S p_m^2 \Delta p$, positive half	1.7653 $\pm$ 0.0198	0.6934 $\pm$ 0.0065

3.2 ^{56}Fe (fig 7, full 0–80 MeV window)

The 16 rows of `fig7_q1p2.dat`, with the per-bin 3D weights ($\Delta p = 40$ MeV/c, p_m at the bin centre): the 2π column summed over all 16 signed bins is the 3D integral N ; the 4π column over all 16 bins is twice that, since the file is symmetric and each $|p_m|$ appears on both sides:

bin	p_m (MeV/c)	$S(p_m)$ (MeV $^{-3}$)	$4\pi S(p_m) p_m^2 \Delta p$	$2\pi S(p_m) p_m^2 \Delta p$
1	−300	9.66999×10^{-9}	0.4375	0.2187
2	−260	2.45150×10^{-8}	0.8330	0.4165
3	−220	6.71838×10^{-8}	1.6345	0.8172
4	−180	1.41263×10^{-7}	2.3006	1.1503
5	−140	2.10452×10^{-7}	2.0734	1.0367
6	−100	2.52322×10^{-7}	1.2683	0.6342
7	−60	2.69957×10^{-7}	0.4885	0.2443
8	−20	3.33847×10^{-7}	0.0671	0.0336
9	+20	3.33847×10^{-7}	0.0671	0.0336
10	+60	2.69957×10^{-7}	0.4885	0.2443
11	+100	2.52322×10^{-7}	1.2683	0.6342
12	+140	2.10452×10^{-7}	2.0734	1.0367
13	+180	1.41263×10^{-7}	2.3006	1.1503
14	+220	6.71838×10^{-8}	1.6345	0.8172
15	+260	2.45150×10^{-8}	0.8330	0.4165
16	+300	9.66999×10^{-9}	0.4375	0.2187
sum, all 16 bins		2.61842×10^{-6}	18.2057	9.1029

quantity	value
$\sum S(p_m)$, plain sum (MeV $^{-3}$)	$2.61842 \times 10^{-6} \pm 2.4 \times 10^{-8}$
$\sum S(p_m) \Delta p$, signed axis (MeV $^{-2}$)	$1.0474 \times 10^{-4} \pm 9.4 \times 10^{-7}$
$4\pi \sum_{p_m > 0} S p_m^2 \Delta p$, positive half	9.1029 $\pm$ 0.0498

Reproduce:

```
pixi run python report/dutta-integral/integrate_dutta.py \
--files fig6_top_q1p2 fig6_bot_q1p2 fig7_q1p2
```